

Hands-on lab on Hadoop Map-Reduce



Objectives

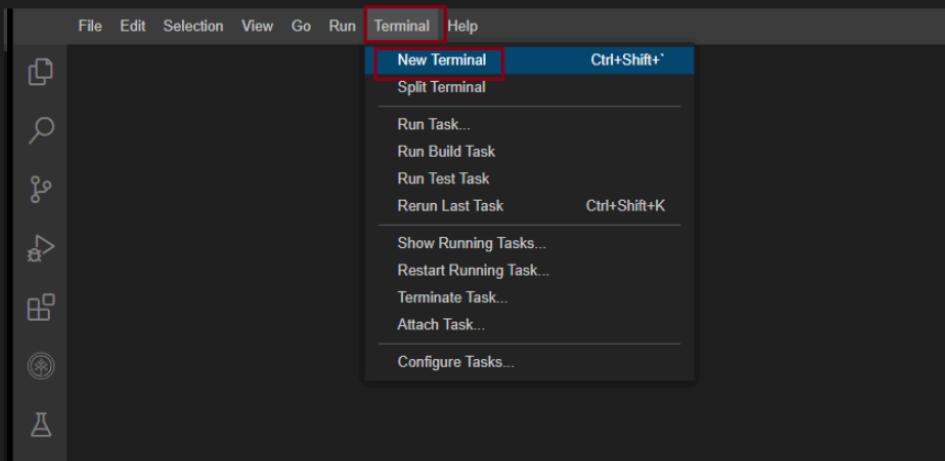
- Run a single-node Hadoop instance
- Perform a word count using Hadoop **Map Reduce**.

Set up Single-Node Hadoop

The steps outlined in this lab use the single-node Hadoop Version 3.3.6. **Hadoop** is most useful when deployed in a fully distributed mode on a large cluster of networked servers sharing a large volume of data. However, for basic understanding, we will configure Hadoop on a single node.

In this lab, we will run the WordCount example with an input text and see how the content of the input file is processed by WordCount.

1. Start a new terminal



2. Download hadoop-3.2.3.tar.gz to your theia environment by running the following command.

```
1 curl https://dlcdn.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz --output hadoop-3.3.6.tar.gz
```

3. Extract the tar file in the currently directory.

```
1 tar -xvf hadoop-3.3.6.tar.gz
```

4. Navigate to the hadoop-3.3.6 directory.

```
1 cd hadoop-3.3.6
```

5. Check the hadoop command to see if it is setup. This will display the usage documentation for the hadoop script.

```
1 bin/hadoop
```

6. Run the following command to download data.txt to your current directory.

```
1 curl https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-BD0225EN-SkillsNetwork/labs/data/data.txt
```

7. Run the Map reduce application for wordcount on data.txt and store the output in /user/root/output

```
1 bin/hadoop jar share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.6.jar wordcount data.txt output
```

This may take some time.

8. Once the word count runs successfully, you can run the following command to see the output file it has generated.

```
1 ls output
```

You should see part-r-00000 with _SUCCESS indicating that the wordcount has been done.

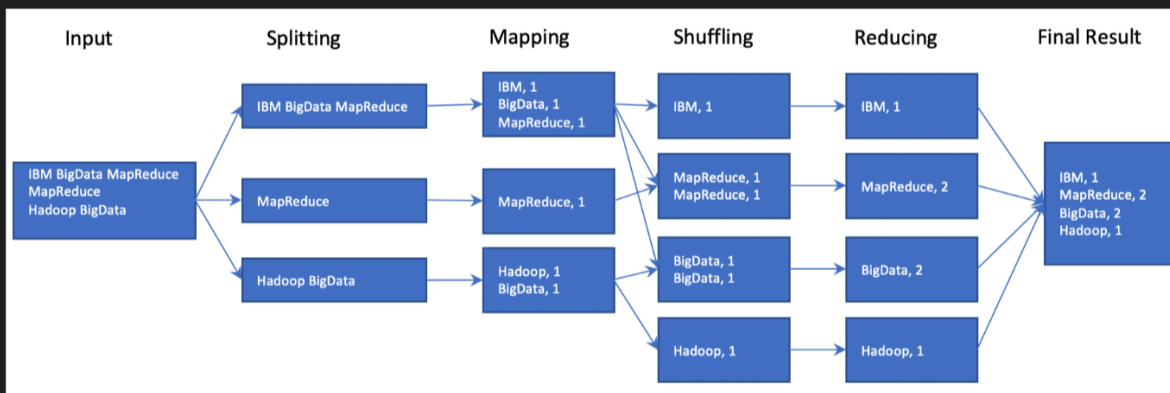
While it is still processing, you may only see '_temporary' listed in the output directory. Wait for a couple of minutes and run the command again till you see output as shown above.

9. Run the following command to see the word count output.

```
1 cat output/part-r-00000
```

```
theia@theiadocker-lavanyas: /home/project/hadoop-3.2.2$ cat output/part-r-00000
BigData 2
Hadoop 1
IBM 1
MapReduce 2
```

The image below shows how the MapReduce wordcount happens.



Practice Lab

1. Do a word count on a file with the following content.

```
1 Italy Venice
2 Italy Pizza
3 Pizza Pasta Gelato
```

▼ Click here for a hint on how to get started

- Delete the data.txt file and output folder

```
1 rm data.txt
```

```
1 rm -rf output
```

▼ Click here for hint on how to create a file to wordcount

Create data.txt with the required content. You may either use the file editor.

▼ Click here for solution on how to do word count on the file

Run the following command

```
1 bin/hadoop jar share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.6.jar wordcount data.txt output
```

▼ Click here for sample output

The output will be as below.

```
root@e4d298bfe26c:/# hdfs dfs -cat /user/root/output/part-r-00000
2021-07-13 05:21:45,467 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
Gelato 1
Italy 2
Pasta 1
Pizza 2
Venice 1
```

Congratulations! You have:

- Deployed Hadoop using Docker
- Copied data into HDFS
- Used MapReduce to do a word count